

Tazim Khan

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EDUCATION

University of Ottawa

Ottawa, ON

BASc Software Engineering (CO-OP)

Sept. 2022 – Dec. 2026

- **Coursework:** Software Design and Architecture, Advanced Programming Concepts With C++, Data Comm. and Networking, Embedded Systems, Applied Machine Learning, Data Structures and Algorithms, Operating Systems

EXPERIENCE

Software Engineer Intern

May 2025 – Present

Natural Resources Canada

Ottawa, ON

- Developed a **Python microservice** that initializes production tracking and triggers entry-point microservices, serving as the starting point for images in the GeoAI pipeline and providing end-to-end traceability across the workflow.
- Built a **Python microservice** that monitors a **PostGIS database** for images needing flatfile creation, extracts geospatial data, generates JSON files, uploads them to **AWS S3**, and updates records for **1000+ images in under 60 seconds**.
- Refactored and updated **10+ microservices** to add new features and address changes in requirements.
- Optimized **GitLab CI/CD workflow** and **Docker configuration** to publish internal Python packages to the GitLab PyPI registry, streamlining dependency management across **20+ microservices** in the pipeline.

Software Engineer Intern

May 2023 – Sept. 2023, Jan. 2024 – Apr. 2024

Canada Revenue Agency

Ottawa, ON

- Developed **20+ reusable Angular components** that were integrated into a shared component library, enabling teams to efficiently digitalize CRA forms.
- Built **10+ Angular pages** to showcase the component library, demonstrating functionality and use cases.
- Implemented **10+ REST APIs** in a **Java Maven** project to validate and transform user inputs for PDF generation.

Development Team Lead

Jan. 2025 – Dec. 2025

Software Engineering Student Association | sesa-aegl.ca | [source](#)

Ottawa, ON

- Leading a team of 5 developers in building and maintaining the SESA website with **Next.js** and **TypeScript**.
- Built backend services for the resources page, centralizing **200+ academic resources** into a structured, searchable repository with **PostgreSQL**, streamlining access and usage for **500+ students**.
- Developed **5 responsive pages** and **20+ reusable components**, ensuring consistency with design specifications.

PROJECTS

Ottawa CrimeLens | *React (TypeScript), Spring Boot (Java), PostgreSQL, Docker, AWS* | ottawacrimelens.ca | [source](#)

- Designed and implemented a geospatial crime ingestion pipeline using **Spring Boot** and **PostgreSQL/PostGIS** to ingest, normalize, and persist municipal crime data for downstream APIs and user-facing applications.
- Deployed scheduled ingestion cron jobs via **AWS ECS Fargate** and **ECR**, triggered by **AWS EventBridge**, with **SNS** alerts reporting job success and failure.
- Built user-facing services by exposing performant spatial query REST APIs and integrating them with an interactive map-based frontend for real-time crime exploration and analysis in Ottawa.

TalentSync | *React (TypeScript), Python, AWS, Docker* | [TalentSync](#) | [source](#)

- Built a Chrome extension that analyzes a user's resume against live job postings by scraping job content in-browser and scoring fit using Llama 3.
- Designed a serverless architecture with **AWS Lambda**, **S3**, and **API Gateway** to enable scalable scraping and inference via Playwright and Groq on the cloud, securing API endpoints with **Google OAuth** and **AWS Cognito**.

TeeChat | *React, Node.js, Express.js, Socket.IO, MongoDB* | teechat.chat | [source](#)

- Developed a real-time chat application with **25 monthly active users**, enabling seamless instant messaging.
- Built backend using **Node.js**, **Express.js**, **MongoDB**, and **Socket.IO** for real-time bidirectional communication.
- Implemented a complete **JWT** auth flow that handles logins, token management, and authorized Socket.IO connections.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, TypeScript, C++, C#, SQL

Frameworks/Libraries: Spring Boot, Flask, FastAPI, Node.js, Express.js, React, Next.js, Angular

Developer Tools: Git, GitHub, GitLab, Docker, AWS, Google Cloud Platform